Course Syllabus: CSS 382

Instructor: Dr. Yusuf Pisan, pisan@uw.edu UW1-260Q (425) 352-3741

Class: Tue/Thu 11:00-1:00 in UW2-240

Office Hours: Tue 10:00-11:00am, Thursday 1:15-2:15pm at UW1-260Q. Email me 24 hours ahead of time to guarantee that I will be in my office, otherwise I might be grabbing coffee or wondering around campus.

Class Discord: invite link

Course Description

Principal ideas and developments in artificial intelligence, such as problem solving, knowledge representation, search, reasoning under uncertainty, learning, and natural language processing.

Course Learning Goals

At the end of this course, students will be able to:

1. Describe what artificial intelligence (AI) means and how machines can process information intelligently;

2. Identify the different fields that comprise AI as well as techniques and heuristics for each area;

3. Create a formal representation of a complex problem; and

4. Write programs to solve complex problems using AI methods in areas such as search, reasoning, natural language processing, games, and robotics.

Textbooks

Lots of books are available for reference, but you do not need to buy any of them.

Russell, Stuart J., and Norvig, Peter. *Artificial Intelligence : a Modern Approach*. 3rd edition, Pearson, 2015. http://aima.cs.berkeley.edu/ and https://www.amazon.com/Artificial-Intelligence-Approach-Stuart-Russell/dp/9332543518

Raschka, S. (2025). Build a large language model (from scratch) ([First edition].). Manning Publications. https://ieeexplore.ieee.org/book/10745358 (e-book from UW library)

Google Colaboratory Examples: Official examples and tutorials from the Google Colab team.

TensorFlow Tutorials: TensorFlow provides a range of tutorials and examples.

Kaggle: Kaggle Notebooks

Machine Learning Mastery

Gymnasium. On reinforcement learning

Gymnasium on GitHub

Fast.ai Courses

Papers with Code

Towards Data Science

Stanford CSS221

Even more AI Courses

1. Anthropic: anthropic.skilljar.com
2. Google: grow.google/ai
3. Meta: ai.meta.com/resources
4. NVIDIA: developer.nvidia.com/training (Best)
5. Microsoft: learn.microsoft.com/training
6. OpenAI: academy.openai.com
7. IBM: skillsbuild.org
8. AWS: skillbuilder.aws
9. DeepLearningAI: deeplearning.ai
10. Hugging Face: huggingface.co/learn
11. FastAI: course.fast.ai
12. Kaggle Learn: kaggle.com/learn
13. Stanford AI: cs231n.stanford.edu
14. MIT OpenCourseWare: ocw.mit.edu
15. Full Stack Deep Learning: fullstackdeeplearning.com
16. DeepMind Resources: https://deepmind.google/
17. OpenAI Cookbook: https://github.com/openai/openai-cookbook

18. Papers With Code: paperswithcode.com
19. AssemblyAI Blog: assemblyai.com/blog
20. Pinecone Learn: https://www.pinecone.io/learn/

Grading

- In-Class Exercises: 20%
- Weekly Projects: 40%
- Group Project: 40%

A scale of 90s (3.5■4.0), 80s (2.5■3.4), 70s (1.5■2.4), 60s (0.5■1.4) is a rough guide. A student who achieves 75% of the possible points will receive a 2.0 grade in this course. Students with 95% and above will receive a 4.0 grade. Scores in between will be interpolated.

In-Class Exercises: You can miss one exercise without penalty. Exercises will be in the form of quizzes or other work and will be completed in class. If you are not physically in class, you will not get any credit. If you expect to miss more than 1 class during the quarter, it will impact your learning and your grade.

Weekly Projects: Reflection, programming or a mix of both. You can submit one and only one project 24 hours late without any penalty. If you are using your 24 hour extension, state that you are using your extension on the "Using Extension" assignment.

Group Project: Will be completed in groups of 2-3 people. You will submit a proposal with milestones and will give updates on your project in class.

You are expected and encouraged to use AI in all your work. However, you must also understand, the code or explanation produced by AI. AI does a great job in "coding", but is not so good in "writing reflections". I want to hear your opinions, not AIs opinions.

Attendance: Attend all classes. You are responsible for all the material covered in class, as well as any announcements including change of due dates or assignment specifications. There will also be graded in-class group exercises to practice problem solving. If you miss a class, I expect you to make-up for it on your own by asking your friends, reviewing the textbook, lecture materials, etc.

Communication: We will use discord as an extension of the classroom. Use a meaningful nickname and act professionally. If your question can be answered publicly by me or by a classmate, post it to discord. Use the office hours for complex issues or topics you are struggling with.

Use your UW email rather than "Canvas Messaging" to communicate directly with me. "Canvas Submission Comments" should only be used to draw the grader's attention to a specific part of your submission.

Problems: If you are having difficulties, come and talk to me. If I don't know about it, I cannot help you. Small problems can be fixed easily early in the quarter, but might become impossible to fix later on.

Course Material: Lecture notes and other material will be posted to Canvas under "Files".

See the School of STEM Course Policies, which covers:

- Academic Integrity
- Access and Accommodations
- Classroom Emergency Preparedness
- For Our Veterans
- Grade of Incomplete
- Inclement Weather

- Parenting Resources
- Religious Accommodations
- Respect for Diversity
- Student Support Services
- Surviving Sexual and Relationship Violence
- Wonder How to Address Faculty?

Course Calendar

Week	Topics	Notes
Week-1 31 Mar/2 Apr	Introduction	Due: Universal Paperclips
Week-2 7/9 Apr	Search	Due: Collaborative Problem Solving

Week-3 14/16 Apr	Multi-Agent Search	Due: Search
Week-4 21/23 Apr	Markov Decision Processes	Due: Multi-Agent Search
Week-5 28/30 Apr	Understanding LLMs Working with Text	Due: Reinforcement Learning
Week-6 5/7 May	Coding Attention Mechanisms Model Construction	Due: Project Proposal
Week-7 12/14 May	Model Optimization Classification and Fine Tuning	Due: Milestone Presentations
Week-8 19/21 May	Conversational Intelligence	Due: Milestone Presentations
Week-9 26/28 May	TBD	Due: Project Deliverables
Week-10 2/4 June	Final project presentations	Due: Final Presentation
Finals week	No final exam (ignore what MyUW says)	